

7 Prenatal testing is used to screen for genetic disorders in developing embryos.

- (a) Phenylketonuria (PKU) is one genetic disorder that is screened for by prenatal testing.

This disorder is inherited in a similar way to cystic fibrosis.

Explain why a healthy couple might choose to have a prenatal test for PKU.

Draw a genetic diagram in your answer.

(4)

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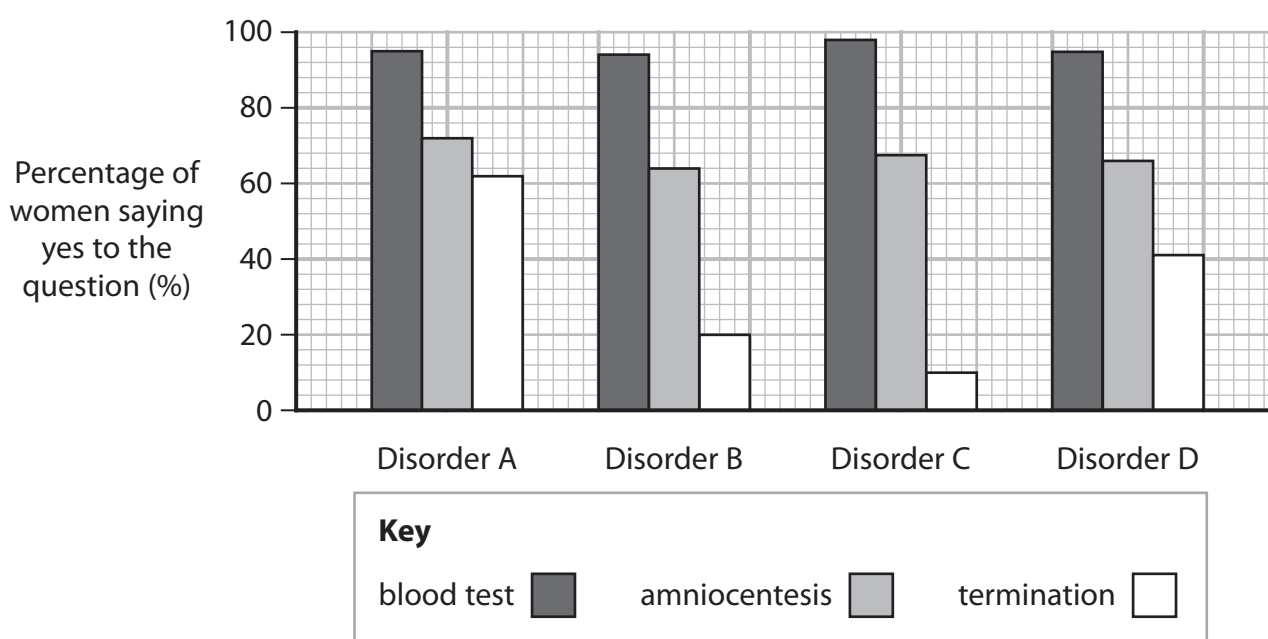


- *(b) A group of women, who had recently given birth to healthy babies, were asked about their opinions on screening for four different genetic disorders, A, B, C and D.

They were asked three questions about each disorder:

- 1 Would they be screened themselves with a blood test?
- 2 Would they have amniocentesis to screen for the genetic disorders?
- 3 Would they have a pregnancy termination (abortion) if an unborn child was found to have the genetic disorder?

The graph shows the percentage of women who said yes to each question for each disorder.



The table gives some information about each of these genetic disorders.

Genetic disorder	Information
A	muscles are weak and movement is difficult symptoms can be managed some types are lethal in childhood
B	treatable does not shorten life expectancy
C	treatable most individuals live into adulthood
D	intellectual disability medical care results in individuals leading healthy lives



Discuss the responses to the three questions.

(6)

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(Total for Question 7 = 10 marks)

